A Lawful Source-Realization Framework for Low-Energy Physical Scales

An "Almost Theory of Everything" Draft with Explicit Seals, Hypotheses, Calculations, and Falsification Targets

Working scientific draft generated from a source-typed research program

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Abstract

This document reconstructs a proposed source-realization framework in the most conservative scientific order available at this stage. It begins with established physics: the constants $c, \hbar, G$, Planck units, the quantum-localization/gravitational-response interface, the Standard Model chiral socket, anomaly constraints, the seesaw rank bound, and ordinary observable transport by renormalization. It then introduces a minimal finite source kernel used as a hypothesis generator, not as a substitute for dynamics. The kernel uses an eight-dimensional phase-space carrier $V_8 = X_4 \oplus P_4$, a three-generation flavor space F_3, a rank-two Majorana realization, normalized traces, and target-dependent phase transport. The central nonstandard branch is deliberately labeled as a source hypothesis: it proposes a small defect parameter

$$S_\star = \frac{1}{8\pi} \frac{2 \cdot 7}{6 \cdot 8 \cdot 9 - 1},$$

a flavor seed $\epsilon = (9S_\star/5)^{1/4}$, and finite phase maps that generate a CKM-scale hierarchy, a charged-lepton C_3 square-root cone, a rank-two neutrino spectrum, and a candidate vacuum boundary relation $\rho_\Lambda = |M_{\nu,ee}|^4$. The document is not presented as a completed derivation of all physical reality from pure mathematics. It is a checkable source-realization model: dimensionless source geometry supplies ratios and phases, while physical realization seals supply dimensional scale, matter sockets, rank conditions, vacuum projection, and observable transport. The final tables list all numerical outputs, reference values, relative residuals, and currently unknown predictions.

Contents

1 Claim discipline

The purpose of this draft is not to claim that all of physics follows from pure geometry. A lawful theory of physical reality must separate at least five source types:

$$\boxed{\text{established theorem} \neq \text{source hypothesis} \neq \text{physical realization seal} \neq \text{observable transport} \neq \text{open wound.}}$$

This separation is central. A dimensionless mathematical structure cannot by itself create an absolute SI or natural-unit scale. A mass formula at a source scale is not automatically a pole mass. A quotient construction for vacuum energy is not automatically a gravitational dynamics. These distinctions are not weaknesses; they are what make the framework scientifically checkable.

Type	Meaning in this document
established physics	Standard physics or standard mathematical fact used without novelty claim.
physical realization seal	A minimal physical input not derived here, such as ℓ_P^2, the Standard Model chiral socket, or a rank-two heavy-neutrino realization.
source hypothesis	A finite-dimensional or arithmetic hypothesis proposed by the framework.
bridge hypothesis	A rule connecting source quantities to physical quantities, not yet derived from a full action or dynamics.
open wound	A precise missing object or theorem.

The lawful maximum form is

$$\boxed{\mathcal{U}_{\text{lawful}} = \mathcal{G}_{\text{source}} + \mathcal{R}_{\text{phys}} + \mathcal{T}_{\text{obs}}.}$$

Here $\mathcal{G}_{\text{source}}$ is a dimensionless grammar of finite carriers, normalized traces, and phase maps. $\mathcal{R}_{\text{phys}}$ contains physical realization seals. $\mathcal{T}_{\text{obs}}$ is the renormalization, threshold, scheme, and pole-mass transport layer needed to compare source values with measured observables.

2 Established constants and the Planck interface

The established physical constants used at the root are

$$c, \qquad \hbar, \qquad G.$$

They define the Planck length, time, and energy:

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}, \qquad t_P = \sqrt{\frac{\hbar G}{c^5}}, \qquad E_P = \sqrt{\frac{\hbar c^5}{G}},$$

with

$$E_P t_P = \hbar.$$

For a probe energy E, define the reduced quantum localization length and a gravitational response length by

$$\lambda_Q(E) = \frac{\hbar c}{E}, \qquad r_G(E) = \frac{GE}{c^4}.$$

Then

$$\lambda_Q(E) r_G(E) = \frac{\hbar c}{E}\frac{GE}{c^4} = \frac{\hbar G}{c^3} = \ell_P^2.$$

Likewise,

$$\tau_Q(E) = \frac{\hbar}{E}, \qquad \tau_G(E) = \frac{GE}{c^5}$$

give

$$\tau_Q(E)\tau_G(E) = t_P^2.$$

At self-duality,

$$\lambda_Q(E_P) = r_G(E_P) = \ell_P, \qquad \tau_Q(E_P) = \tau_G(E_P) = t_P.$$

This is established dimensional physics. It says that the Planck area is the product of quantum localization and gravitational response. It does not derive the numerical value of ℓ_P^2 from pure dimensionless mathematics.

2.1 Dimensional no-go

If a source grammar $\mathcal{G}_{\text{source}}$ is dimensionless, then every pure function $f(\mathcal{G}_{\text{source}})$ is dimensionless. Since

$$[\ell_P^2] = L^2,$$

one has

$$\mathcal{G}_{\text{source}} \not\Rightarrow \ell_P^2$$

unless a dimensionful physical realization seal is admitted. This is a no-go boundary, not a failure. In this framework ℓ_P^2, or equivalently $M_{\text{P}}^{\text{red}}$, is a physical realization seal.

We use the reduced Planck normalization

$$A_{2\pi} = 8\pi\ell_P^2, \qquad {M_{\text{P}}^{\text{red}}}^2 = \frac{1}{8\pi\ell_P^2}.$$

The numerical calculations use the CODATA-style reduced Planck value

$$M_{\text{P}}^{\text{red}} = 2.4353234600842885 \times 10^{18}\ \text{GeV}.$$

3 The phase-space carrier and the diagonal measurement sheet

The source carrier is not bare spacetime alone. It is the doubled phase-space-type carrier

$$V_8 = X_4 \oplus P_4,$$

where X_4 is an event-space carrier and P_4 is the conjugate response carrier. Let

$$\Omega = \sum_{\mu=0}^{3} dp_\mu \wedge dx^\mu.$$

The diagonal measurement sheet is

$$D_4 = \text{span}\{x_0 + p_0,\ x_1 + p_1,\ x_2 + p_2,\ x_3 + p_3\} \subset X_4 \oplus P_4.$$

In binary form it is the row pattern

$$\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1. \end{array}$$

For $d_\mu = x_\mu + p_\mu$, one has

$$\Omega(d_\mu, d_\nu) = 0,$$

so D_4 is Lagrangian. For the time-energy socket,

$$x^0 = ct, \qquad p_0 = \frac{E}{c},$$

hence

$$p_0 x^0 = Et.$$

At the quantum action cell $Et = \hbar$. Thus the diagonal sheet is not an arbitrary diagram: it is the finite-dimensional place where event coordinates and conjugate response form action.

4 Minimal physical realization inputs

The model uses the following minimal physical realization inputs.

4.1 Metric-area seal

The absolute area scale ℓ_P^2 is not derived from dimensionless source geometry. It enters as the metric-area seal.

4.2 Standard Model chiral socket

The chiral matter socket

$$Q, \quad u^c, \quad d^c, \quad L, \quad e^c, \quad \nu^c, \quad H$$

is treated as a physical realization input unless a future finite-matter theorem derives it. Once it is assumed, hypercharge is forced by standard anomaly and Yukawa constraints.

Let

$$Y(Q) = q, \quad Y(u^c) = u, \quad Y(d^c) = d, \quad Y(L) = \ell, \quad Y(e^c) = e, \quad Y(\nu^c) = n, \quad Y(H) = h.$$

Yukawa invariance gives

$$q + h + u = 0, \quad q - h + d = 0, \quad \ell - h + e = 0, \quad \ell + h + n = 0.$$

The anomaly constraints include

$$2q + u + d = 0, \qquad 3q + \ell = 0,$$

$$6q + 3u + 3d + 2\ell + e + n = 0,$$

and

$$6q^3 + 3u^3 + 3d^3 + 2\ell^3 + e^3 + n^3 = 0.$$

The solution is

$$\ell = -3q, \qquad h = 3q, \qquad u = -4q, \qquad d = 2q, \qquad e = 6q, \qquad n = 0.$$

With $h = 1/2$,

$$Y(Q) = \frac{1}{6}, \quad Y(u^c) = -\frac{2}{3}, \quad Y(d^c) = \frac{1}{3}, \quad Y(L) = -\frac{1}{2}, \quad Y(e^c) = 1, \quad Y(\nu^c) = 0.$$

This is an established conditional result: if the Standard Model chiral socket exists, its hypercharges are essentially forced. It does not prove the socket itself.

4.3 Minimal rank-two seesaw realization

A minimal two-source Type-I seesaw uses

$$N_R \in \mathbb{C}^2, \qquad Y_\nu \in \mathrm{Mat}_{2\times 3}(\mathbb{C}), \qquad M_R \in \mathrm{Mat}_{2\times 2}(\mathbb{C}).$$

Then

$$M_\nu = -\frac{v^2}{2} Y_\nu^T M_R^{-1} Y_\nu$$

satisfies

$$\operatorname{rank} M_\nu \le 2.$$

For rank-two Y_ν,

$$\operatorname{rank} M_\nu = 2, \qquad m_1 = 0, \qquad m_2, m_3 \neq 0.$$

The rank-two realization is a lawful physical seal/theorem pair. It is not forced by pure geometry here, but once admitted it has strong downstream consequences.

5 Finite source kernel

Let

$$F_3 \simeq \mathbb{C}^3, \qquad \dim F_3 = 3, \qquad \dim \mathrm{End}(F_3) = 9.$$

Let $V_8 = X_4 \oplus P_4$, $\dim V_8 = 8$. Fix one observer/action line $\ell_0 \subset V_8$ and define

$$C_7 = V_8/\ell_0, \qquad \dim C_7 = 7.$$

In four event dimensions,

$$\dim \Lambda^2 X_4 = \binom{4}{2} = 6.$$

The finite defect chamber is

$$\Lambda^2 X_4 \otimes V_8 \otimes \mathrm{End}(F_3),$$

with dimension

$$6 \cdot 8 \cdot 9 = 432.$$

Removing the scalar identity gives the traceless chamber dimension

$$432 - 1 = 431.$$

The rank-two Majorana defect support is

$$\mathrm{rank}\, M_\nu \cdot \dim C_7 = 2 \cdot 7 = 14.$$

Define

$$L = \frac{1}{8\pi}.$$

The central small source parameter is then

$$\boxed{S_\star = L\frac{2 \cdot 7}{6 \cdot 8 \cdot 9 - 1} = \frac{14}{431}\frac{1}{8\pi}.}$$

Numerically,

$$S_\star = 0.001292441533.$$

Define the flavor seed

$$\boxed{\epsilon = \left(\frac{9}{5}S_\star\right)^{1/4}.}$$

Numerically,

$$\epsilon = 0.2196194769.$$

The factor $9/5 = 3 \cdot 3/5$ is interpreted as color multiplicity times inverse $5/3$ hypercharge normalization. This is a source hypothesis, not established Standard Model dynamics.

6 Normalized trace phase transport

For a finite projector P on a finite carrier W, define

$$\tau(P; W) = \frac{\mathrm{Tr}\, P}{\dim W}.$$

The framework uses a target-dependent phase transport rule:

$$\Theta(\tau, T) = \begin{cases} 2\pi\tau, & T = U(1) \text{ complex phase,} \\ \pi\tau, & T = \mathbb{RP} \text{ projective orientation,} \\ \tau, & T = C_3 \text{ square-root cone coordinate.} \end{cases}$$

This rule is a bridge hypothesis. It is not yet derived from a fundamental action.

6.1 The 3/8 phase

Let $P_{\rm sp}$ project onto the three spatial event directions inside V_8. Then

$$\frac{\operatorname{Tr} P_{\rm sp}}{\dim V_8} = \frac{3}{8}.$$

For projective orientation,

$$\delta_q = \pi \frac{3}{8} = \frac{3\pi}{8}.$$

6.2 The 2/9 cone coordinate

The rank-two Majorana density in 3×3 flavor endomorphism space is

$$\theta_\ell = \frac{\operatorname{rank} M_\nu}{\dim \operatorname{End}(F_3)} = \frac{2}{9}.$$

This is used as the cone coordinate of the charged-lepton square-root mass vector.

6.3 The 2/3 cyclic phase and Koide cone

Let $C_3 = \langle \omega : \omega^3 = 1 \rangle$. The nontrivial $U(1)$ phase is

$$\Phi = \frac{2\pi}{3}.$$

For charged leptons, define

$$q_\ell = (\sqrt{m_\tau}, \sqrt{m_e}, \sqrt{m_\mu}) = q_{\mathbf{1}} + q_{\mathbf{2}},$$

where $q_{\mathbf{1}} \parallel (1,1,1)$ and $q_{\mathbf{2}} \perp (1,1,1)$. If

$$\|q_{\mathbf{1}}\| = \|q_{\mathbf{2}}\|,$$

then

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}.$$

Equivalently,

$$q_k = A \left[1 + \sqrt{2} \cos \left(\theta_\ell + \frac{2\pi k}{3} \right) \right], \qquad k = 0, 1, 2,$$

automatically lies on the Koide cone.

7 Electroweak and heavy-anchor source formulas

The electroweak source action is

$$\mathcal{A}_{\rm EW} = -12\pi + \frac{\sqrt{3}}{2} + 2S_\star + 148 S_\star^2.$$

The source-level Higgs vacuum expectation value is

$$v = M_{\rm P}^{\rm red} e^{\mathcal{A}_{\rm EW}}.$$

The Higgs quartic source bridge is

$$\lambda = \frac{3}{8}(1 + L) \left(\frac{1}{3} - S_\star \right).$$

The tree-level Higgs mass bridge is

$$m_H = v\sqrt{2\lambda}.$$

Using the CODATA reduced Planck seal, the script gives

$$v = 246.2523746 \text{ GeV}, \qquad \lambda = 0.1294696422, \qquad m_H = 125.3081732 \text{ GeV}.$$

This is not claimed as a pole-mass theorem. A loop and threshold transport layer is required.

The heavy Yukawa source exponents are

$$A_t = L - 5S_\star + 138S_\star^2,$$

$$A_b = \frac{4\pi}{3} - 56S_\star + 106S_\star^2,$$

$$A_\tau = \frac{4\pi}{3} + \frac{3}{10} + \frac{7}{72} - S_\star + \frac{99}{2}S_\star^2.$$

Then

$$y_t = e^{-A_t}, \qquad y_b = e^{-A_b}, \qquad y_\tau = e^{-A_\tau}.$$

Numerically,

$$y_t = 0.9669997481, \qquad y_b = 0.01629999396, \qquad y_\tau = 0.01020576349.$$

8 Flavor construction

8.1 CKM orientation

Define

$$s_{12}^q = \epsilon\sqrt{1+\epsilon^2}, \qquad s_{23}^q = \frac{\sqrt{3}}{2}\epsilon^2, \qquad s_{13}^q = \frac{1}{2\sqrt{2}}\epsilon^3,$$

and

$$\delta_q = \frac{3\pi}{8}.$$

The standard CKM parameterization gives

$$|V_{\text{CKM}}| = \begin{pmatrix} 0.974386 & 0.224852 & 0.00374514 \\ 0.224716 & 0.973529 & 0.0417705 \\ 0.00867751 & 0.0410305 & 0.99912 \end{pmatrix}$$

and

$$J_q = 3.1637606586 \times 10^{-5}.$$

This is a source-level CKM orientation. Exact quark pole masses are not claimed because they require RG transport.

8.2 Charged-lepton cone

The charged-lepton cone uses

$$\theta_\ell = \frac{2}{9}$$

and

$$q_k = A\left[1 + \sqrt{2}\cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right)\right].$$

With the source-level τ value from the heavy anchor,

$$m_\tau = 1777.096162 \text{ MeV},$$

the cone predicts

$$m_e = 0.5110309253 \text{ MeV}, \qquad m_\mu = 105.6660259 \text{ MeV}.$$

The residuals relative to measured charged-lepton masses are listed in Section 11. They are at the 10^{-4} level before any observable transport correction.

8.3 Rank-two neutrino branch

The dominant heavy Majorana scale is

$$M_{R3} = (\sqrt{2\pi} + 49S_\star + 90S_\star^2)\sqrt{vM_{\mathrm{P}}^{\mathrm{red}}},$$

and

$$M_{R2} = M_{R3}\frac{e^{-4\pi/3}}{4L + 10S_\star}.$$

Numerically,

$$M_{R2} = 5.5465470951 \times 10^9 \text{ GeV}, \qquad M_{R3} = 6.2939016877 \times 10^{10} \text{ GeV}.$$

The light spectrum is

$$m_1 = 0,$$

$$m_3 = \frac{v^2 e^{-2A_\tau}}{2M_{R3}}, \qquad m_2 = (4L + 10S_\star)m_3.$$

Numerically,

$$m_2 = 0.008634370516 \text{ eV}, \qquad m_3 = 0.05017667775 \text{ eV}.$$

8.4 PMNS branch

The PMNS angle skeleton is

$$\theta_{12} = \frac{\pi}{6} + 48S_\star,$$

$$\theta_{13} = 4L - 7S_\star + 4S_\star^2,$$

$$\theta_{23}^- = \frac{\pi}{4} - 48S_\star, \qquad \theta_{23}^+ = \frac{\pi}{4} + 48S_\star.$$

The current branch chooses the lower octant:

$$\theta_{23} = \theta_{23}^-.$$

The phase kernel gives

$$\delta_{\mathrm{PMNS}} = 4\delta_q = \frac{3\pi}{2},$$

and the Majorana boundary condition gives

$$\alpha_{\mathrm{eff}} + 2\delta_{\mathrm{PMNS}} = \frac{2\pi}{3},$$

so

$$\alpha_{\mathrm{eff}} = \frac{5\pi}{3}.$$

The PMNS modulus is

$$|U_{\mathrm{PMNS}}| = \begin{pmatrix} 0.823989 & 0.546513 & 0.149551 \\ 0.422451 & 0.627067 & 0.654464 \\ 0.377595 & 0.555077 & 0.741155 \end{pmatrix}$$

9 Vacuum boundary candidate

For a rank-two Majorana neutrino matrix, define

$$M_\nu = U^*_{\text{PMNS}}\,\text{diag}(0, m_2, m_3) U^\dagger_{\text{PMNS}}.$$

The electron-boundary Majorana scalar is

$$M_{\nu,ee} = e^T M_\nu e.$$

Equivalently,

$$M_{\nu,ee} = m_2 s_{12}^2 c_{13}^2 e^{i\alpha} + m_3 s_{13}^2 e^{-2i\delta}.$$

With

$$\Phi_{\beta\beta} = \alpha + 2\delta = \frac{2\pi}{3},$$

one has

$$|M_{\nu,ee}|^2 = a^2 + b^2 - ab,$$

where

$$a = m_2 s_{12}^2 c_{13}^2, \qquad b = m_3 s_{13}^2.$$

The calculation gives

$$|M_{\nu,ee}| = 2.239632931 \text{ meV}.$$

The vacuum boundary candidate is

$$\boxed{\rho_\Lambda^{\text{cand}} = |M_{\nu,ee}|^4.}$$

Numerically,

$$\rho_\Lambda^{\text{cand}} = 2.5159811233 \times 10^{-11} \text{ eV}^4, \qquad \Lambda^{\text{cand}} = 1.0894846938 \times 10^{-52} \text{ m}^{-2}.$$

9.1 Source quotient status

A formal quotient model can be written as

$$C_0^{src} = \langle B_0, B_m, B_H, B_g, D\rangle,$$

$$C_1^{src} = \langle R_0, R_m, R_H, R_g\rangle, \qquad \partial_{src} R_i = B_i,$$

so

$$H_0^{src} = C_0^{src}/\operatorname{im}\partial_{src} \cong \langle D\rangle,$$

with

$$D = \left[|M_{\nu,ee}|^4, \,(e, \nu, \Delta L = 2, C_3, \Phi = 2\pi/3)\right].$$

This formal quotient is mathematically consistent. The physical assertion that gravity implements this quotient is a physical realization seal/bridge hypothesis, not yet an established theorem.

10 Observable transport

A source-level prediction is not automatically a measured pole value. A lawful comparison requires

$$\mathcal{T}_{\text{obs}} : \mathcal{L}_{\text{source}} \to \mathcal{L}_{\text{measured}}.$$

For Yukawa values this has the schematic form

$$Y_f^{src}(\Lambda) \to Y_f^{\overline{\text{MS}}}(\mu) \to m_f^{pole}.$$

For the Higgs quartic,

$$\lambda^{src}(\Lambda) \to \lambda^{\overline{\text{MS}}}(\mu) \to m_H^{pole}.$$

The transport layer must specify

$$\Lambda_{\text{source}}, \quad g_1, g_2, g_3, \quad \text{scheme}, \quad \text{thresholds}, \quad \text{loop order}, \quad \text{uncertainty propagation}.$$

This paper does not claim that this layer is complete. The error table should therefore be read as a source-level test, not as a final precision electroweak fit.

11 Calculation and error tables

The following values are generated by the accompanying Python script. Reference values are representative central values from CODATA, PDG, NuFIT, and Planck-2018 style sources. The relative residual is

$$\frac{\text{model} - \text{reference}}{\text{reference}}.$$

The status column specifies whether the residual is considered a direct failure, a source-level residual requiring transport, or a prediction.

Table 1: Numerical outputs and reference comparisons.

Quantity	Model value	Reference value	Relative residual	Status
v [GeV]	246.2524	246.2197	1.3291×10^{-4}	source-level; CODATA Planck seal
m_H [GeV]	125.3082	125.2	8.6400×10^{-4}	tree-level bridge; loop transport required
m_τ [MeV]	1777.096	1776.86	1.3291×10^{-4}	source-level edge value
m_e [MeV]	0.5110309	0.5109989	6.2574×10^{-5}	Koide-cone source value; residual may be transport
m_μ [MeV]	105.666	105.6584	7.2407×10^{-5}	Koide-cone source value; residual may be transport
$\sin^2\theta_{12}$	0.3055099	0.307	-0.004854	NuFIT comparison; source skeleton
$\sin^2\theta_{13}$	0.02236562	0.02203	0.01523	NuFIT comparison; source skeleton
Δm^2_{21} [eV^2]	7.4552354×10^{-5}	7.4900000×10^{-5}	-0.004641	NuFIT comparison
Δm^2_{31} [eV^2]	0.002517699	0.002513	0.00187	NuFIT comparison
Λ [m^{-2}]	$1.0894847 \times 10^{-52}$	$1.0909105 \times 10^{-52}$	-0.001307	Planck-2018 comparison; cosmological uncertainty dominates
$\rho_\Lambda^{1/4}$ [meV]	2.239633	2.240365	-3.2691×10^{-4}	dark-energy fourth-root scale
$\lvert V_{us}\rvert$	0.224852	0.2243	0.002461	representative CKM comparison
$\lvert V_{cb}\rvert$	0.04177046	0.0415	0.006517	representative CKM comparison
$\lvert V_{ub}\rvert$	0.003745136	0.00382	-0.0196	representative CKM comparison

11.1 Acceptability rubric

Case	Interpretation
Established derivation	No novelty claimed; used as physics background.
Exact source identity	Algebraically exact inside the proposed finite source kernel.
Source-level residual 10^{-4}-10^{-2}	Not a final measurement fit. It requires observable transport or becomes a falsification target.
Anchor	A quantity used to set a sector scale, not counted as an independent prediction.
Prediction	Not yet decisively known or not directly measured. Future data can kill it.

12 Predictions and future kill tests

Table 2: Predictions or sharp future tests.

Target	Model value	How it can be tested
Lightest neutrino mass	$m_1 = 0$	Can be falsified by absolute-mass data incompatible with rank two.
Sum of neutrino masses	$\sum m_\nu = 0.05881105\,\mathrm{eV}$	Within current cosmological upper bounds; future tightening tests it.
Atmospheric octant branch	$\theta_{23} = 41.445531^\circ$; $\sin^2\theta_{23} = 0.438122$	Killed if the upper octant is established decisively.
Leptonic CP phase	$\delta_{\mathrm{PMNS}} = 3\pi/2$	Future long-baseline experiments can test/exclude.
Effective Majorana mass	$m_{\beta\beta} = 2.239633\,\mathrm{meV}$	Below near-term $0\nu\beta\beta$ reach; sharp long-term prediction.
Heavy Majorana scales	$M_{R2} = 5.546547 \times 10^9\,\mathrm{GeV}$, $M_{R3} = 6.293902 \times 10^{10}\,\mathrm{GeV}$	Model-internal high-scale prediction; indirectly testable.
Vacuum-boundary relation	$\rho_\Lambda = \lvert M_{\nu,ee}\rvert^4$	Killed if improved neutrino and cosmology data break the relation.
CKM source phase	$\delta_q = 3\pi/8$	Already close to CKM fits; refined global fits can stress it.

13 What is solved, what is not solved

13.1 Stabilized pieces

1. The Planck interface:
$$\lambda_Q r_G = \ell_P^2, \qquad \tau_Q \tau_G = t_P^2.$$
2. The dimensional no-go: dimensionless source geometry cannot derive ℓ_P^2.
3. The finite defect density:
$$S_\star = \frac{1}{8\pi}\frac{2\cdot 7}{6\cdot 8\cdot 9 - 1}.$$
4. The flavor seed:
$$\epsilon = \left(\frac{9}{5}S_\star\right)^{1/4}.$$

5. The phase kernel:
$$\delta_q = \frac{3\pi}{8}, \quad \theta_\ell = \frac{2}{9}, \quad \Phi_{\beta\beta} = \frac{2\pi}{3}, \quad \delta_{\mathrm{PMNS}} = \frac{3\pi}{2}.$$
6. The charged-lepton C_3 square-root cone and the rank-two Majorana branch.

13.2 Explicit remaining seals and wounds

1. ℓ_P^2 remains a physical metric-area seal.
2. The Standard Model chiral socket is assumed, though hypercharge follows once it is assumed.
3. The rank-two seesaw is a minimal physical realization; a deeper origin would strengthen the model.
4. The phase-transport rule is structured but still a bridge rule.
5. The vacuum quotient is formal; gravitational implementation is not derived.
6. The observable transport layer is mandatory and incomplete.

14 Maximum honest statement

The framework is not a completed pure-geometric Theory of Everything. Its lawful claim is weaker and more checkable:

$$\boxed{\text{dimensionless source geometry constrains ratios and phases;}}$$

$$\boxed{\text{physical realization seals instantiate scale, sectors, and rank;}}$$

$$\boxed{\text{observable transport maps source quantities to measurements.}}$$

Under this architecture, the model gives a compact source-realization account of the electroweak scale, Higgs quartic bridge, heavy Yukawa anchors, CKM hierarchy, charged-lepton cone, rank-two neutrino spectrum, PMNS branch, and a dark-energy boundary candidate. Its scientific value depends on whether the frozen rules survive improved flavor data, neutrino absolute-mass data, cosmological measurements, and a completed RG/pole-mass transport layer.

A Reproducibility

All numerical values in the tables are generated by:

`lawful_toe_calculations.py`.

The script writes machine-readable data to:

`lawful_toe_build/calculation_data.json`.

The calculation uses no fitted update after the source kernel is frozen.

B References

References

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